Instructions from NYU for submitting to REU:
Please submit your personal statement below and include a description of any research experience you have. Your personal statement should give us a glimpse into the most important things/events/circumstances that have made you the person you are. In particular, the most compelling statements show that the person has talent, enthusiasm for science, and the ability to overcome challenges, which can be personal or professional. Your statement should be around 500-1000 words (equivalent to 1-2 pages, single-spaced). Please note that your personal statement will not be saved until you navigate to the next page. We recommend starting your personal statement in a separate document and copying and pasting your response below in order to avoid losing your work.

Graces submission:
Finding your purpose is not an easy task, but during the grueling labor of fieldwork, and frustrating components of lab work I found myself. Asking scientific questions and learning the interesting world of biology feels like the interworking gears that drive me. Conducting research is something that I will do for the rest of my life. My goal is to earn my Ph.D. in genomics and study coevolution. In my professional career, I plan to work at a biotechnology company where I will develop novel methods to analyze genomic data including whole genome sequencing data and tissue-specific transcriptomes. This will allow my work to expand the range of questions that researchers in both industry and academia can answer.

I began to develop these goals when I started in the Snakes, Spiders, Scorpions, Toxins, INformatics, and Genomics (SSSTING) Laboratory. I started in March of 2021 with Dr. Jason Strickland, having been drawn there by an interest in toxin and protein analysis. During my time thus far, I have become the first person to generate a venom gland transcriptome for the Common Garter Snake with RNA-seq approaches using Illumina while working on my Honors Thesis: "Understanding Venom Transcriptomics of the Rear-fanged Snake *Thamnophis sirtalis*". I started the project from the ground up, by completing fieldwork, lab work, and bioinformatics. I fell in love with the aspect of completing a difficult project and seeing an answer in the form of ATCGs on a computer screen. After my initial introduction to the bioinformatic process with my project on Garter Snakes, I was eager to continue learning. In collaboration with Dr. Strickland and Dr. Meghan Hermance, I started as a paid research assistant in August of 2022 working on a project titled "A Ticking Time Bomb: Testing how viral infection modifies salivary gland gene expression in an invasive tick during feeding". Through this, I am learning how to incorporate whole genome sequence data to understand changes in genomic architecture that lead to differential gene expression. To have the ability to work on a completely different project, with different goals, while still utilizing my knowledge of genetics and bioinformatics is so humbling, and eye-opening. The process makes me want to see what other questions and problems genetics can be used to answer.

Undergraduate research has also allowed me to become more involved on campus. The Biology Department and Honors College have both asked me to speak about my experiences, and students in my classes frequently ask me about how to get involved in research. This involvement allowed me to develop a community of students and faculty in my department and on campus as I have gotten to know more people. I look forward to walking into the biology building where the faculty and students say “hello” and update me on the interesting developments of their research. I always was able to find a friend or mentor who could advise me on difficulties I came across in my research. Additionally, I developed the desire to always help my classmates. I want to share the support I feel with others in need of it. I assisted the Honors College in starting a panel full of juniors and seniors that aim to help incoming freshmen with beginning research, answering questions, and providing guidance. Additionally, my involvement in Marine Science Student Association and Biology Student Association has helped me make connections with peers and offer them guidance in research. My involvement in my lab and student organizations has provided me with fantastic leadership skills that I am excited to utilize in my professional career, and other opportunities as they come, such as the NSF REU program.

I knew I wanted to pursue an REU at the Simons Foundation when I began reading student testimonials. The growth and experience they had during their time at the Simons Foundation connected with me. While I have been fortunate enough to have undergraduate research experience, I still want to keep growing. Unfortunately, at my university, we do not have enough diverse lab programs that do genetic work to provide the challenge I seek or to gain a variety of skills. The work being conducted at the Simons Foundation is exactly the program I can see myself thriving in. I am highly interested in the work of Dr. Joy Bergelson. Her focus on coevolution and the use of genetics to better manage agriculture, as well as her study on microbiomes and plants, aligns with my interests and provides opportunities for growth and development. However, I am also curious about other work being conducted like that of Dr. Gloria Coruzzi. Her expertise in plant genomics would facilitate my development as a student, geneticist, and researcher. The opportunities at this program are endless and will facilitate an environment to help me grow and hone my skills in a way I will not receive on my own.

Thank you for your time and consideration.

Grace Coppinger